

AI&SD Lab Course

Basic Info

Communication:

- Weekly meetings (virtual)
- Email: s13kwija@uni-bonn.de
- Discord: ktirta

Timeline: TBD

Workload:

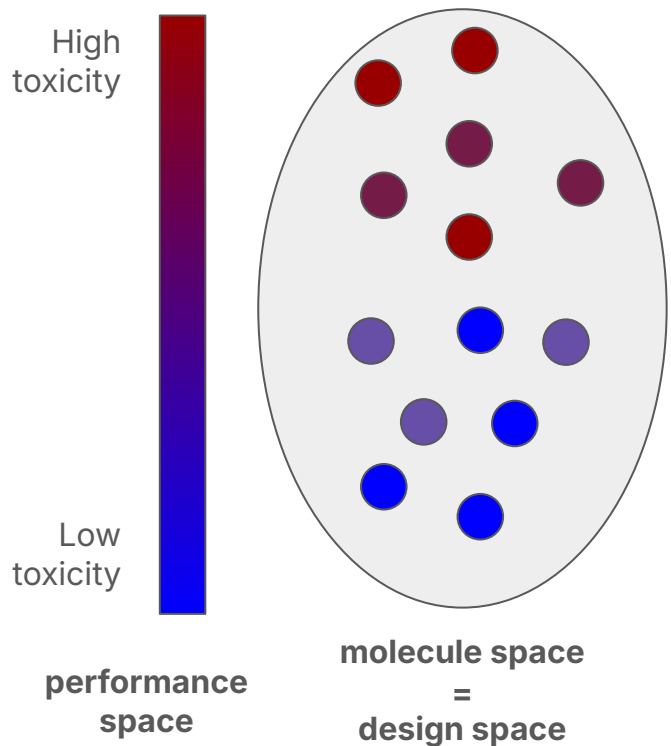
- Literature review
- Implementation

Deliverables:

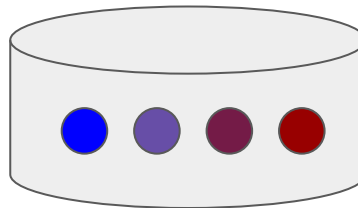
- Presentation
- Report

AI for Ranking-based Optimization

AI for Ranking-based Optimization



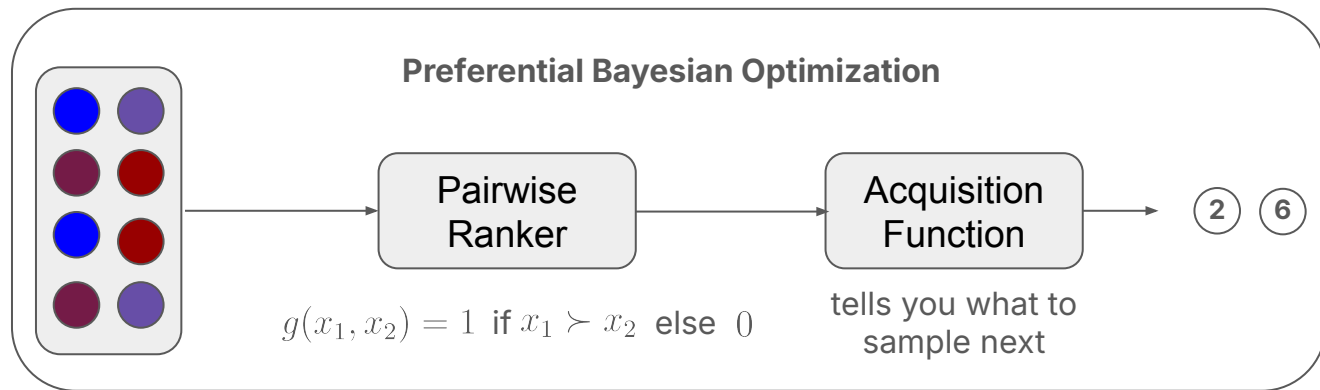
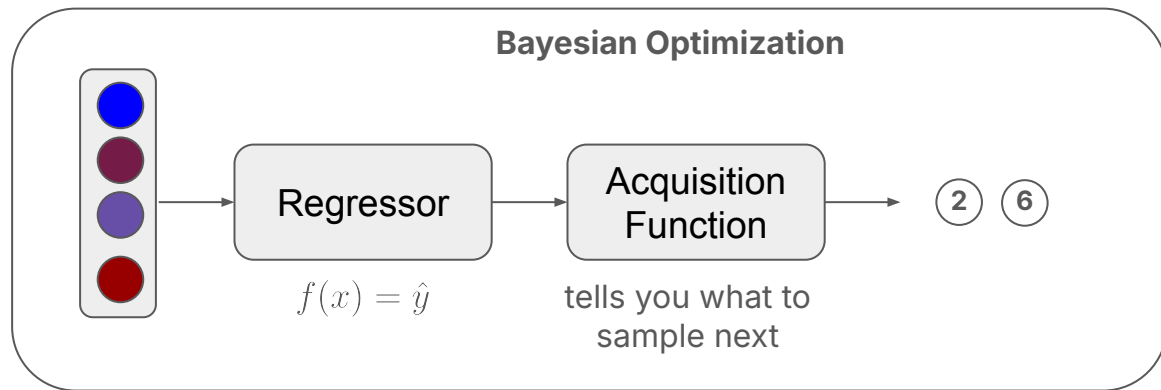
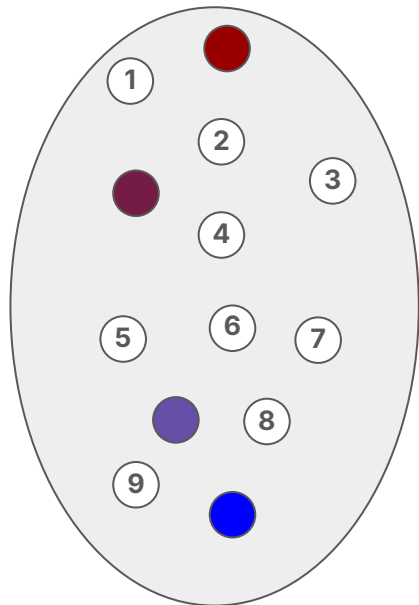
- Experimental validation is slow and expensive
- Scalar labels (e.g., toxicity values) are scarce



Dataset. Maybe we only have 4 labels.

Given a limited experimental budget, how can we find the best performing item? (e.g., least toxic molecule)

AI for Ranking-based Optimization



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1. Find a (quantitative) problem that you like. (e.g., molecule solubility, airfoil shape).

Criteria:

- a. 1-D problem, quantitative problem. We can quantify the performance of a design with a single scalar number.
 - b. We have computational tools to verify the performance of a new design.
2. Implement and evaluate LLM as,
 - a. A regression surrogate, i.e., predicting the absolute performance of each design
 - b. A pairwise ranker surrogate, i.e., predicting the relative performance between design A and design B (is A better than B).
 3. Implement and evaluate optimization framework:
 - a. LLM as a direct optimizer
 - b. Bayesian Optimization with LLM as a surrogate regressor
 - c. Preferential Bayesian Optimization with LLM as a surrogate pairwise ranker

AI for Ranking-based Optimization

Relevant Literatures:

1. Shahriari, Bobak, et al. "Taking the human out of the loop: A review of Bayesian optimization." Proceedings of the IEEE 104.1 (2015): 148-175.
2. González, Javier, et al. "Preferential bayesian optimization." International Conference on Machine Learning. PMLR, 2017.
3. Cai, Chengkun, et al. "Bayesian optimization for controlled image editing via llms." Findings of the Association for Computational Linguistics: ACL 2025. 2025.
4. Jablonka, Kevin Maik, et al. "Leveraging large language models for predictive chemistry." Nature Machine Intelligence 6.2 (2024): 161-169.
5. Liu, Chenxi, Selena Ling, and Alec Jacobson. "GimmBO: Interactive Generative Image Model Merging via Bayesian Optimization." arXiv preprint arXiv:2601.18585 (2026).